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Chun-Hsiung Chiang

National Kaohsiung First University of Science of Technology, u9828904@nkfust.edu.tw

Shih-Wei Chou

National Kaohsiung First University of Science of Technology, swchou@nkfust.edu.tw

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UNDERSTANDING SOFTWARE-AS-A-SERVICE PERFORMANCE—A DYNAMIC CAPABILITY PERSPECTIVE

Chun-Hsiung Chiang, Institute of management, National Kaohsiung First University of
Science of Technology, Taiwan, R.O.C., u9828904@nkfust.edu.tw

Shih-Wei Chou, Department of information management, National Kaohsiung First
University of Science of Technology, Taiwan, R.O.C., swchou@nkfust.edu.tw

Abstract

How to increase a client's capability through outsourcing remains a problem. This paper draws on strategic management literature and the relational view to develop a theoretical model that explains the relationships between collaboration, agility, and outsourcing performance in software-as-a-service (SaaS) context. Collaboration are characterized as knowledge sharing and process alignment between a supplier and its client, agility as a supplier's sensing agility and responding agility. This study also investigates the moderating effect of environmental turbulence on the relationships between agility and performance. The proposed hypotheses are largely supported by the empirical data from 215 firms. The results show that SaaS performance is affected by both sensing agility and responding ability, which, in turn, are impacted by collaboration between a supplier and its client. Finally, we discuss the implications of our results.

Key words: dynamic capability, outsourcing outcomes, software-as-a-service, collaboration.

1 INTRODUCTION

SaaS, referring to on-demand software applications delivered as a service over Internet, is becoming an important model that provides users and firms with Internet-based access to resources and expertise (Benlian & Hess 2011; Benlian et al. 2012). SaaS attracts its clients by benefits including IT cost reductions, operational elasticity, and leveraging suppliers' capability and resources. The growth rate of SaaS revenues is significant (Petty 2006).

However, not everyone is positive about SaaS's feasibility and capability. It has been reported that the failure rate of SaaS is high. The notable drivers of this phenomenon include low-quality customer support, inability to handle process dependence, failure to increase clients' performance, and security issues (Benlian et al. 2009; Benlian et al. 2012). SaaS is deemed as success, if it is able to help client firms improve performance. Improving a client firm's dynamic capabilities motivates it for outsourcing in general and SaaS in particular, because dynamic capabilities reflect the ability to adapt to and perform well in rapidly changing environments (e.g., entering new market segments and finding opportunities for innovation and competitive action) (Bardhan et al. 2010; Rai & Tang 2010; Roberts & Grover 2012). Thus, there is significant motivation to better understand how and why a client's dynamic capabilities can be improved.

Studies on outsourcing have suggested that a strategic perspective should be employed combining organizational and contextual factors (Goo et al. 2007; Quinn 1999). Accordingly, research has considered various antecedents to outsourcing, including organizational characteristics (e.g., knowledge acquisition, strategic importance of IT activity, interaction processes) and contextual factors (e.g., relational norms and task environment)(Goo et al. 2007; Han et al. 2008; Mani et al. 2010). While prior research on outsourcing has examined the broader influences of capability (clients' or vendors' capability) on performance (Bharadwaj et al. 2010; Han et al. 2008), there is a lack of research that explores how a client's dynamic capabilities are affected by suppliers' capability (conceptualized as customer agility), collaboration (leveraging interfirm resources between a supplier and its client), and environmental turbulence.

Due to the gap in understanding dynamic capabilities in SaaS, this study aims to investigate the effect of suppliers' customer agility on the client's dynamic capabilities and to examine how resources of interorganizational relationship can be exploited to affect customer agility. Building on the relational view and organizational agility literature, (Dove 2001; Dyer & Singh 1998; Overby et al. 2006), we develop the research model. Our results extend current theory by providing managerial insights on the necessary capability either from the vendor or from the interfirm resources to enhance a client's dynamic capability.

2 THEORETICAL FRAMEWORK

2.1 Literature Review

SaaS refers to an on-demand software delivery for (on-premise) commercial off-the-shelf application development. SaaS uses multi-tenant architecture to share IT infrastructure and IS applications efficiently to its clients, leading to high economics of scale. Besides, multi-tenant architecture gives suppliers more control over future development. As a result, clients have more expectations about SaaS services and performance (e.g., reliable and frequent updates, and customer agility) (Benlian et al. 2009; Lu & Ramamurthy 2011; Roberts & Grover 2012).

Extant research has focused on outsourcing decisions and outsourcing outcomes (Dibbern et al. 2004; Lacity et al. 2010). Prior work on dynamic capabilities shows that action efficacy reflects a firm's awareness, motivation, and action that affect competitive action (Chen 1996; Sambamurthy et al. 2003). Action efficacy refers to the extent to which a firm has the ability to sense and respond to opportunities and threats, and take effective action (e.g., innovation in products and services) (Roberts & Grover 2012). Research on outsourcing points out that increasing a client's action efficacy has been viewed as an importance measure of outsourcing success (outcomes), but its formation in SaaS context remains unclear (Bharadwaj et al. 2010; Goo et al. 2007, 2009; Han et al. 2008).

Studies on outsourcing and SaaS have drawn on different economic and organization theories to explain how outsourcing performance can be enhanced. Bharadwaj et al. (2010) found that suppliers' process and IT capability and clients' outsourcing management capability were positively associated with outsourcing outcomes. Others emphasize the importance of collaboration between suppliers and clients in affecting outsourcing relationships and outsourcing performance (Han et al. 2008). Still, others suggest that outsourcing performance relies on strategic management and relationship management, including knowledge acquisition and learning, process alignment, unique services, flexibility to change contractual/ functional aspects of outsourcing tasks, agility in rapidly coping with market or demand changes) (Benlian et al. 2012; Goo et al. 2007; Goo & Huang 2008; Rai and Tang 2010; Sambamurthy et al. 2003).

Among these antecedents, both relational value and a supplier's capability (e.g., agility) have been highlighted, because they play a key role in affecting outsourcing performance in general and SaaS action efficacy in particular (Benlian et al. 2012; Bharadwaj et al. 2010; Goo et al. 2007, 2009). While useful, these studies pay little attention to the influence of dynamic capabilities on outsourcing performance. Enhancing a client's competitive position motivates it to be involved in an outsourcing relationship. Besides, suppliers' dynamic capabilities help their client handle extreme changes and respond quickly to opportunities for innovation and competitive action (Roberts & Grover 2012; Saraf et al. 2007). Dynamic capabilities and outsourcing relationships become particularly important in SaaS. This is because of limited customization and increased client expectations on suppliers' service (e.g., responsiveness) that make enhancement of outsourcing performance difficult (Benlian & Hess 2011).

Dynamic capabilities have been examined from different perspectives, including knowledge creation, process execution, competitive dynamics perspective, the relational view, and the resource-based view (Dyer & Singh 1998; Han et al. 2008; Lu & Ramamurthy 2011; Overby et al. 2006; Rai & Tang 2010; Rai et al. 2012; Roberts & Grover 2012). For example, Lu and Ramamurthy (2011) found that flexible IT planning processes, IT innovations, and acquiring new knowledge affected a firm's dynamic capabilities, conceptualized as ability to respond to changes and operational adjustments. Others characterize dynamic capabilities as action efficacy that meets customer needs (Roberts & Grover 2012). They also suggest that customer-based knowledge and operational process execution play a key role in action efficacy. Few studies consider the influence of dynamic capabilities on outsourcing in SaaS. Studies on outsourcing show that gaining competitive advantages plays a key role in a client's willingness to outsource and has been viewed as a measure of outsourcing success (Han et al. 2008; Lacity et al. 2010; Rai & Tang 2010).

2.2 Hypotheses Development

We use the relational view to explain the influence of collaboration on customer agility. The relational view (Dyer & Singh 1998) identifies four dimensions or relational value: *relationship-specific assets, knowledge-sharing routines, complementary capabilities, and effective governance*.

We propose a collaboration-agility-efficacy framework (Dyer & Singh 1998; Eisenhardt & Martin 2000; Lacity et al. 2010), in which we argue that the extent to which a client is able to improve its action efficacy depends on suppliers' customer agility. This agility, in turn, is influenced by the relational value created by the collaboration between a supplier and its client.

Prior work has conceptualized customer agility as sensing capability and responding ability. In SaaS, sensing capability reflects a supplier's ability to sensing changing client needs, and client-based innovation. This capability also helps the supplier collect a variety of information to identify external changes and understand how to rapidly cope with market changes (Benlian et al. 2012; Lu & Ramamurthy 2011; Sambamurthy et al. 2003). Responding capability in SaaS captures a supplier's ability to quickly implement its planned activities and respond to changes in its client's product or service needs (Roberts & Grover 2012).

Sensing capability provides information related to market development, customer needs, and competitor responses. Thus, this capability is more likely to increase awareness of competitive environment and opportunities. Sensing capability also enhances motivation to take action because of understanding about market changes and competitors. Sensing capability boosts confidence in taking effective action due to understanding about market changes and innovation. In SaaS, given limited customization and different client expectations regarding system performance, a supplier's sensing capability reflect that it better understands its client's needs and has the ability to provide needed services to improve competitive performance. We thus hypothesize that sensing capability positively affects action efficacy (H1).

H1: Sensing capability positively affects action efficacy.

Studies state that a firm with responding capability should be able to understand business environment and react effectively to changing market conditions driven by services that are needed to its client (Broadbent et al. 1999; Cho et al. 1996). Others report that responding capability reflects a firm's intention to make moves against competitors (Rai & Tang 2010; Roberts & Grover 2012). In SaaS, given a client's reliance on its supplier's capability in future development (Benlian & Hess 2011; Benlian et al. 2012), the client is likely to believe that its supplier can provide useful information related to environmental and unanticipated change, and help the client deal with changing market conditions. This also motivates the client to take action.

H2: Responding capability positively affects action efficacy.

Environmental turbulence refers to general conditions of uncertainty and emerges from three key resources—(1) market turbulence: rate of change in the composition of clients and their preference, (2) competitive intensity: the number of competitors in the field, and (3) technological change: the frequency of IT breakthroughs (Jap 2001).

A supplier's agility becomes more important when its client's environment becomes more turbulent (Eisenhardt & Martin 2000; Teece et al. 1997). This is because a client's reliance on external resources (e.g., suppliers' capability or agility) increases compared to a less turbulent environment. A client's competitive performance relies more on its suppliers' capability and agility when environmental turbulence increases (Lacity et al. 2010; Roberts & Grover 2012). We hypothesize that the relationships posited in H1 and H2 will be stronger when environmental turbulence is high and weaker when turbulence is low.

H3: Environmental turbulence strengthens the relationship between sensing capability and action efficacy.**H4: Environmental turbulence strengthens the relationship between responding capability and action efficacy.**

A firm's ability to sense opportunities relies primarily on its ability to create and deepen knowledge because sensing new opportunities requires the firm's learning, scanning, and interpreting activities to understand the features of innovation and opportunities (Broadbent et al. 1999; Overby et al. 2006; Weill et al. 2002). Knowledge also serves as the key driver of responding capability as understanding about customer preferences and constraints, and marketing plans helps the firm develop a comprehensive and viable plan (Jaworski & Kohli 1993; Lu & Ramamurthy 2011; Zhang & Sharifi 2007).

Process alignment reflects a firm's ability to coordinate interdependent activities and optimize operations with its partners (Rai & Tang 2010). Prior work on managing interfirm processes reports that a firm's process alignment helps the coordination and execution of its operational processes, which, in turn, enable responses to opportunities (Roberts & Grover 2012). Besides, well-coordinated

operational processes also implies the increase of information flow and knowledge sharing, and effective IT-business joint decision making, leading to sensing agility (Chan et al. 2006; Teece et al. 1997).

The aforementioned literature shows the importance of process alignment and knowledge sharing in improving collaboration for interfirm business processes in general and outsourcing in particular. In SaaS, due to constraints on customization and different client expectations on performance, collaboration between suppliers and clients becomes more critical to relational value. Thus, we anticipate that a supplier will have more agility if it collaborates more closely with its client.

H5a: Knowledge sharing positively affects suppliers' sensing capability.

H5b: Knowledge sharing positively affects suppliers' responding capability.

H6a: Process alignment positively affects suppliers' sensing capability.

H6b: Process alignment positively affects suppliers' responding capability.

Figure 1 lists the research model.

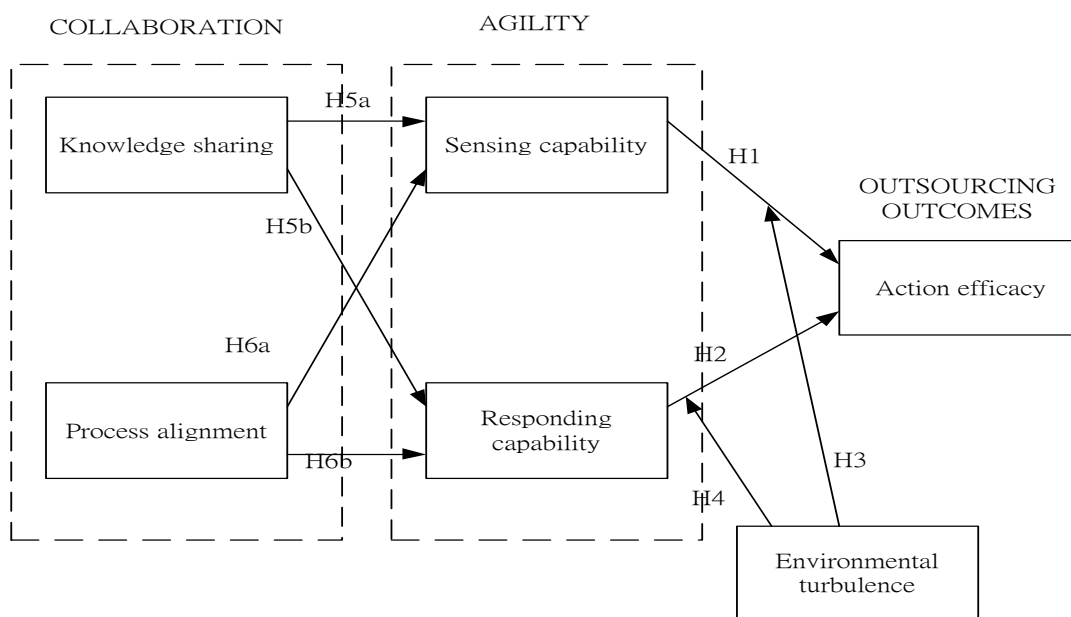


Figure 1. Research model

3 RESEARCH METHODOLOGY AND ANALYSIS

We identified 800 firms with SaaS experience through the assistance of Market Intelligence & Consulting Department under the institute for Information Industry in Taiwan. Senior IT managers were chosen as the key informants because of their knowledge about outsourcing and cloud computing applications. Of the 600 distributed surveys, 264 responses were received and 49 responses

were discarded due to missing data. Thus, 215 responses were retained (26% response rate) in the final analysis (Table 1).

		Frequency	Percentage (%)
Industry sector	Computers and software	29	14
	Manufacturing	39	18
	Banking/finance	22	10
	Publishing/communication	30	14
	Medicine/healthy	42	20
	Service	23	11
	Government/public	16	7
	other	14	6
Respondent title	CEO, CIO	23	11
	IS manager (applications)	62	29
	Business operation managers	77	36
	Other managers	53	24
Annual revenues (NT\$)	Less than 2 billion	42	19
	2-3 billion	81	38
	3-5 billion	65	30
	Over 5 billion	27	13
Number of employees	Less than 500	78	36
	501-1000	85	40
	1001-3000	39	18
	Over 3000	13	6

Table 1. Sample profile (N = 215)

All the survey items were measured using a seven-point Likert scale ranging from 1 to 7, with 7 “completely agree” and 1 “completely disagree.” To evaluate the common method biases (CMV) that may occur in questionnaire-based research, we used the following strategies. First, we used technique of counterbalancing question order and psychological separation of measurement (Podsakoff et al. 2003). To enable psychological separation, we designed two versions of the questionnaire, including different scale items, changed scale endpoints for independent and dependent variables, and controlled scale length. The questionnaire was pretested to reduce the possible flaws and weakness. 40 MIS students with knowledge about SaaS and an expert panel tested the questionnaire. Second, we used Harmon’s one-factor test to evaluate the CMV. As expected, six factors with eigenvalues > 1 were

extracted and collectively accounted for 58% of the variance in the data, with the first factor accounting for 24% of the variance. Thus, we conclude that CMV is not significant in our data.

This study used partial least squares (PLS) analysis. Factor loading of each item was used to evaluate individual item reliability. A high loading suggests that the shared variance between constructs and its measurement is higher than error variance (Chin et al. 2003). A factor loading higher than 0.7 is treated as high reliability and a factor loading less than 0.5 should be dropped (Hair et al. 1998). The loadings of all indicators are larger than 0.65.

Construct reliability was assessed based on Cronbach's alpha. Discriminant validity is assured when (1) cross-loadings show all items have a higher loading in the defined construct than in any other construct, (2) correlations between pairs of constructs is less than 0.9, and (3) the square root of average variance extracted (AVE) is larger than the correlation between constructs (Fornell & Larcker 1981). Table 2 shows that Cronbach's alphas were greater than 0.7 and AVEs were larger than 0.5, indicating high internal consistency. Table 3 suggests that these conditions hold, showing that these constructs had adequate discriminant and convergent validity.

Constructs	# of items	Composite Reliability	Average Variance Extracted	Cronbach Alpha
Knowledge Sharing(KS)	3	0.863	0.679	0.768
Process Alignment (PA)	5	0.828	0.548	0.725
Sensing Capability(SC)	5	0.915	0.682	0.883
Responding Capability(RC)	6	0.864	0.515	0.812
Action Efficacy(AE)	5	0.845	0.522	0.770
Environmental Turbulence(ET)	3	0.815	0.595	0.759

Table 2. Reliability and average variance extracted

	Mean	S.D.	KS	PA	SC	RC	AE	ET
KS	4.767	0.973	0.824					
PA	4.737	0.874	0.294	0.740				
SC	4.658	1.002	0.229	0.536	0.826			
RC	4.871	0.960	0.487	0.449	0.427	0.718		
AE	5.018	0.888	0.303	0.354	0.291	0.428	0.722	
ET	4.851	0.961	0.294	0.257	0.236	0.378	0.435	0.771

Table 3. Correlation between construct (The shaded numbers in the diagonal row are square roots of the average variance extracted (AVE))

H5a and H5b predicted the effects of knowledge sharing on agility. From model 1 in Table 4, the results showed that knowledge sharing had a significant impact on responding capability ($\beta=0.389$, $p<0.01$) but the influence on sensing capability is not significant ($\beta=0.078$, $p=n.s.$). Hence, H5b was supported, but H5a was not. H6a and H6b assessed the effects of process alignment on agility (see model 1 in Table 4). Process alignment had significant effects on both sensing capability $\beta=0.513$, $p<0.01$ and responding capability ($\beta=0.334$, $p<0.01$). Thus, H6a and H6b were confirmed.

Model	1(all)	2	3
Dependent variable: action efficacy			
Sensing capability (SC)	0.132*		
Responding capability (RC)	0.376***		
SC* environmental turbulence		0.136*	
RC* environmental turbulence			0.223**
R ²	0.201	0.214	0.249
Differenced R ²		0.013	0.048
f ²		0.017	0.064
Test of differenced R ²		3.587*	13.504***
Dependent variable: sensing capability			
Knowledge sharing	0.078		
Process alignment	0.513***		
R ²	0.293		
Dependent variable: responding capability			
Knowledge sharing	0.389***		
Process alignment	0.334***		
R ²	0.339		

Table 4. Structural model results ($F(0.1, 1, 211)=2.729$; $F(0.05, 1, 211)=3.886$; $F(0.01,1,211)=6.757$; * $p<0.1$; ** $p<0.05$; *** $p<0.01$)

H1 and H2 predicted the effects of agility on action efficacy. Both sensing capability ($\beta=0.132$, $p<0.1$) and responding capability ($\beta=0.376$, $p<0.01$) had a significant influence on action efficacy, explaining 20% of the variance of action efficacy (see model 1 in Table 4).

We evaluated the moderating effects by comparing the difference between the main effect and the moderating effect models (Chin et al. 2003). The R-square (R_1^2) of the main effect was obtained, including the independent variable, moderator, and dependent variable only. Next, we calculated the R-square (R_2^2) of the moderating effect model, including all the variables in the main model and the interaction term in the model. The interaction terms were obtained by adding the product of each indicator in the independent variable and the moderator. Then, we measured an estimated effect size

of f^2 from $(R_2^2 - R_1^2) / (1 - R_2^2)$ and a pseudo F-value by multiplying f^2 with $(n-k-1)$, where n is the sample size and k is the number of independent variables in the regression equation. Based on the results in Table 4, both H5 (model 2) and H6 (model 3) were supported.

4 DISCUSSION

Our results show interesting implications for SaaS research. First, our results generally show that collaboration does affect the extent to which a supplier senses and responds to changing client needs and client-based innovation. The proposed collaboration-agility-efficacy framework confirms that a provider's agility plays a key role in leveraging the resources (i.e. relational value through collaboration with the client) by SaaS and affects outsourcing outcomes.

Second, research has been somewhat inconsistent on how much agility is influenced by IT capability or IT-enabled services (Lu and Ramamurthy 201; Roberts and Grover 2012), with some research emphasizing IT features (e.g. IT innovations, integrating strategic planning and IT planning) and others fail to substantiate a relationship between features and performance (Overby et al. 2006; Sambamurthy et al. 2003). Our results suggest that a supplier's agility is influenced by its ability to leveraging the resources embedded in the relationship with the client. Besides, our findings show that the influence of agility on SaaS performance (or action efficacy) is contingent on the environmental turbulence. This implies that achieving agility requires not only the collaboration between the client and the supplier by knowledge sharing and process alignment, but paying special attention to the client environment.

This study has two limitations. First, several other factors salient to agility are not discussed in this paper. For example, agility may be evaluated from other perspectives such as IS integration (e.g. internal and external IS integration). Future work may consider the impact of IS integration on agility and SaaS performance. Second, the use of a cross-sectional survey suffers from the limitation of drawing definitive conclusions about causal relationships among constructs. Future work may use an in-depth process-oriented approach to understand underlying causes explaining the effect of collaboration on agility.

References

- Bardhan, I. R., Demirkan, H., Kannan, P. K., Kauffman, R. J., and Sougstad, R. (2010). An interdisciplinary perspective of IT services management and service science. *Journal of MIS*, 26(4), 13-64.
- Benlian, A., and Hess, T. (2011). Opportunities and risks of software-as-a-service: Findings from a survey of IT executives. *Decision Support Systems*, 52, 232-246.
- Benlian, A., Hess, T., and Buxmann, P. (2009). Drivers of SaaS-adoption: An empirical study of different application types. *Business & Information Systems Engineering*, 1(5), 357-369.

- Benlian, A., Koufaris, M., and Hess, T. (2012). Service quality in software-as-a-service: Developing the SaaS-Qual measure and examining its role in usage continuance. *Journal of MIS*, 28(3), 85-126.
- Bharadwaj, S. S., Saxena, K., and Halemane, M. (2010). Building a successful relationship in business process outsourcing: An exploratory study. *European Journal of Information Systems*, 19, 168-180.
- Broadbent, M., Weill, P., and Neo, B. S. (1999). Strategic context and patterns of IT infrastructure capability. *Journal of Strategic Information Systems*, 8(2), 157-187.
- Chan, Y. E., Sabherwal, R., and Thatcher, J. B. (2006). Antecedents and outcomes of strategic IS alignment. *IEEE Transaction on Engineering Management*, 51(3), 27-47.
- Chen, M. J. (1996). Competitor analysis and interfirm rivalry: Toward a theoretical integration. *Academy of Management Review*, 21(1), 100-134.
- Chin, W. W., Marcolin, B. L., and Newsted, P. R. (2003). A partial least squares latent variable modeling approach for measuring interaction effects--Results from a Monte Carlo simulation study and an electronic-mail emotion/adoption study. *Information Systems Research*, 14(2), 189-217.
- Cho, H., Jung, M., and Kim, M. (1996). Enabling technologies of agile manufacturing and its related activities in Korea. *Computers and Industry Engineering*, 30(3), 323-335.
- Dibbern, J., Goles, T., Hirschheim, R., and Bandula, J. (2004). Information systems outsourcing: A survey and analysis of the literature. *Database for Advances in Information Systems*, 34(4), 6-102.
- Dove, R. (2001). *Response agility: The language, structure, and culture of the agile enterprise*, New York: John Wiley & Sons.
- Dyer, J. and Singh, H. (1998). The relational view: Cooperative strategy and sources of interorganizational competitive advantage. *Academy of Management Review*, 23(4), 660-679.
- Eisenhardt, K. M. and Martin, J. A. (2000). Dynamic capabilities: What are they? *Strategic Management Journal*, 21, (10-11), 1105-1121.
- Fornell, C., and Larcker, D. F. (1981). Structural equation models with unobservable variables and measurement errors. *Journal of Marketing Research*, 18(2), 39-50.
- Goo, J., Kishore, R., Nam, K., Rao, H. R., and Song, Y. (2007). An investigation of factors that influence the duration of IT outsourcing relationships. *Decision Support Systems*, 42, 2107-2125.
- Goo, J., Huang, C. D. (2008). Facilitating relational governance through service level agreements in IT outsourcing: An application of the commit-trust theory. *Decision Support Systems*, 46, 216-232.
- Goo, J., Kishore, R., and Rao, H. R. (2009). The role of service level agreements in relational management of information technology outsourcing: An empirical study. *MIS Quarterly*, 33(1), 119-145.
- Hair, J. F., Anderson, R. E., Tatham, R. L., and Black, W. C. (1998). *Multivariate data analysis* (5th ed.), Prentice Hall, Englewood Cliffs, NJ.
- Han, H. S., Lee, J. N., Seo, Y. W. (2008). Analyzing the impact of a firm's capability on outsourcing success: A process perspective. *Information & Management*, 45, 31-42.

- Hitt, M. A., Keats, B. W., and DeMarie, S. M. (1998). Navigating in the new competitive landscape: Building strategic flexibility and competitive advantage in the 21st century. *Academy of Management Executive*, 12(4), 22-42.
- Jap, S. D. (2001). Perspectives on joint competitive advantage in buyer-supplier relationships. *Internet Journal Research Marketing*, 18(1), 19-35.
- Jaworski, B. J. and Kohli, A. K. (1993). Market orientation: Antecedents and consequences. *Journal of Marketing*, 57(3), 53-70.
- Lacity, M. C., Khan, S., Yan, A., and Willcocks, L. P. (2010). A review of the IT outsourcing empirical literature and future research directions. *Journal of Information Technology*, 25, 395-433.
- Lu, Y., and Ramamurthy, K. (2011). Understanding the link between information technology capability and organizational agility: An empirical examination. *MIS Quarterly*, 35(4), 931-954.
- Mani, D., Barua, A., and Whinston, A. (2010). An empirical analysis of the impact of information capabilities design on business process outsourcing performance. *MIS Quarterly*, 34(1), 39-62.
- Overby, E., Bharadwaj, A., and Sambamurthy, V. (2006). Enterprise agility and the enabling role of information technology. *European Journal of Information Systems*, 15(2), 120-131.
- Pettey, C. (2006). Gartner says 25 percent of new business software will be delivered as software as a service by 2011. Gartner, Stamford, CT.
- Podsakoff, P., MacKenzie, S., Lee, J., and Podsakoff, N. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychological*, 88(5), 879-903.
- Rai, A., Pavlou, P., Im, G., and Du, S. (2012). Interfirm IT capability profiles and communications for cocreating relational value: Evidence from the logistics industry. *MIS Quarterly*, 36(1), 233-262.
- Rai, A., and Tang, X. (2010). Leveraging IT capabilities and competitive process capability for the management of interorganizational relationship portfolios. *Information Systems Research*, 21(3), 516-542.
- Roberts, N., and Grover, V. (2012). Leveraging information technology infrastructure to facilitate a firm's customer agility and competitive activity: An empirical investigation. *Journal of MIS*, 28(4), 231-269.
- Zhang, Z. and Sharifi, H. (2007). Towards theory building in agile manufacturing strategy: A taxonomical approach. *IEEE Transactions on Engineering Management*, 54(2), 351-370.